

PARAS BANSAL

📁 Portfolio | ✉️ heyparasbansal@gmail.com | ☎️ +91-7889099620 | in LinkedIn | 🐙 GitHub

Education

Indian Institute of Information Technology, Allahabad

B.Tech in Information Technology; CGPA: **9.00**

Aug 2024 – May 2028

Prayagraj, Uttar Pradesh

Golden Earth Global School, Sangrur

Class 12 (CBSE); Percentage: **94.6%**

Apr 2023 – Mar 2024

Sangrur, Punjab

Golden Earth Global School, Sangrur

Class 10 (CBSE); Percentage: **97%**

Apr 2021 – Mar 2022

Sangrur, Punjab

Competitive Programming

CodeChef (**4-Star**, Rating 1820)

| Codeforces (**Expert**, Rating 1710)

| LeetCode (**Knight**, Rating 1914)

Projects

Pseudo2CPP

Feb 2026 – Mar 2026

C++, NLP, Compiler Design, Data Structures & Algorithms

GitHub

- Implemented an **NLP-driven C++ transpiler** converting English algorithmic queries into optimized **Data Structures & Algorithms (DSA)** implementations using **50+ reusable templates**.
- Designed a **semantic keyword engine** analyzing **100+ tokens** to classify algorithms with **90%+ accuracy**.
- Built modular **parsing, semantic analysis, and code generation pipelines** enabling automated competitive programming boilerplate with **O(1)** template retrieval.

Energy-Aware Beam-ACO Routing

Oct 2025 – Dec 2025

C++, Metaheuristics, Algorithm Design, Optimization

GitHub

- Implemented a hybrid **Ant Colony Optimization (ACO) + Beam Search** routing algorithm in C++, reducing path-planning cost by **15%**.
- Integrated **Beam Search top-K pruning** to limit exploration space, improving convergence speed by **40%**.
- Designed a physics-based **energy cost model** using velocity and drag, improving routing efficiency by **15%**.

Samadhaan

Jun 2025 – Aug 2025

MERN Stack, REST APIs, System Design, MongoDB

GitHub | Live Demo

- Developed a scalable **MERN-based grievance management platform** supporting **12+ civic complaint categories** with real-time status tracking.
- Built and documented **15+ RESTful APIs** using **Node.js** and **Express.js** with **JWT authentication** and role-based access control.
- Optimized backend performance using **MongoDB indexing**, reducing API latency by **20%** and improving UI load time by **25%**.

Technical Skills

Languages: C++ (STL), C, Java, JavaScript (ES6+), Python (Intermediate), HTML5, CSS3, SQL

Technologies/Frameworks: Git, GitHub, Linux, React.js, Node.js, Tailwind CSS, NumPy, MongoDB, MySQL

Core Concepts: Data Structures & Algorithms (DSA), Object-Oriented Programming (OOP), System Design, Database Management Systems (DBMS), Operating Systems, Computer Networks, Theory of Computation

Achievements

- Global Top 5% (Expert) on Codeforces** with rating **1710**; ranked **830 / 91,904+** active users in India with best contest rank **451** in Codeforces Round 1066 (Div.1 & 2).
- Achieved **Global Rank 93** in CodeChef Starters 224 and **Global Rank 38** in LeetCode Weekly Contest 484, competing among thousands of participants worldwide.
- Solved **1100+** problems across diverse **Data Structures & Algorithms (DSA)** topics on Codeforces, CodeChef, and LeetCode.
- JEE Main 2024: **AIR 3341** (**Top 0.2%** among 1.4M candidates); JEE Advanced 2024: **AIR 9232**.

Experience

Academic Mentor

2023 – Present

Physics Wallah

- Mentored **1000+ JEE aspirants** on problem-solving strategies, doubt resolution, and exam preparation.

Head Boy

May 2021 – Apr 2022

Golden Earth Global School

- Led a student **council of 35 members** to organize **10+ school events**, improving school-wide communication.